

Claim Payment Variance Analyzer: Healthcare Revenue Integrity using Apache Spark

IST 718: Big Data Analytics

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1. Project Overview

1.1 Introduction

The United States healthcare system operates on a complex web of reimbursement models, payer contracts, and regulatory coding guidelines. Within this framework, hospitals and healthcare providers face a critical financial challenge: **payment variance**. Payment variance is defined as the difference between the expected contractual reimbursement for a medical service and the actual payment received from the insurer¹. While this variance can occasionally result in overpayment, it most commonly manifests as underpayment, representing a significant revenue leak for providers.

This project, titled the "Claim Payment Variance Analyzer," seeks to address this issue by leveraging Big Data technologies. Using **Apache Spark**, our team successfully processed large-scale Medicare claim data to build a data-driven framework capable of predicting hospital claim charges and analyzing payment deviations. By identifying claims where the actual payment significantly diverges from the predicted expected amount, this tool empowers Revenue Cycle Management (RCM) stakeholders to proactively recover lost revenue.

1.2 Business Problem and Justification

The financial impact of payment variance is substantial. According to the Medical Group Management Association, insurers underpay providers by an average of 7% to 11%. Furthermore, research by the American Hospital Association revealed that in 2020, 67% of hospitals received Medicare payments below their actual costs. These discrepancies accumulate rapidly; an Advisory Board report estimates that the average hospital loses approximately \$22 million annually due to underpayments.

These losses are rarely intentional but rather the result of complex payer contracts, systemic claim coding errors, and administrative inefficiencies. Manual auditing of every claim is impossible due to the sheer volume of data. Therefore, an automated, machine-learning-based solution is required to "flag" high-risk claims for review. This project solves that problem by providing a predictive baseline for what a claim should pay, allowing for the automatic detection of revenue leaks.

2. Project Objectives

Our group established three primary goals to guide the development of the Claim Payment Variance Analyzer. These goals were designed to satisfy both the scientific requirements of the course and the operational needs of the healthcare business case.

2.1 Prediction Goal

The primary technical objective was to build a regression model using **PySpark** to predict the `predicted_payment` amount for inpatient hospital claims. The model utilizes key claim

characteristics-specifically Diagnosis-Related Group (DRG) weights, Length of Stay (LOS), and patient demographics-to establish an "expected" payment baseline. This prediction serves as the ground truth against which actual payments are measured.

2.2 Inference Goal

Beyond simple prediction, we sought to understand the drivers of healthcare costs. Our inference goal was to quantify the relationship between clinical complexity (measured by DRG weight) and financial reimbursement. Specifically, we aimed to identify patterns or anomalies that indicate underpayment or overpayment trends across different providers and regions. By analyzing the coefficients and feature importance of our Spark model, we can infer which variables (e.g., a patient's state of residence or chronic condition status) has the most significant impact on payment variance.

2.3 Deviation Analysis (Operational Goal)

The ultimate business utility of this project lies in deviation analysis. We defined a strict logic for operationalizing our predictions:

- **Variance Calculation:** Variance is calculated as the percentage difference between the actual payment and the predicted payment.
- **Flagging Logic:** Any claim with a variance lower than -10% is flagged as a "Potential Underpayment".
- **Visualization:** The system generates output (dashboards) that allows stakeholders to view underpayment patterns by DRG, hospital, or region.

3. Data Description

To simulate a real-world environment while maintaining data privacy, we utilized the **Medicare DE-SynPUF (Data Entrepreneurs' Synthetic Public Use Files)**. This dataset is provided by the Centers for Medicare & Medicaid Services (CMS) and mimics the structure and statistical characteristics of actual Medicare claims. The data is fully relational, requiring significant join operations-a task well-suited for Spark's distributed processing capabilities.

We integrated data from three distinct sources:

3.1 Beneficiary Summary Data

This dataset acts as the "Master Patient Index," containing demographic and enrollment information for Medicare beneficiaries.

- **Volume:** 116,352 rows and 32 columns.
- **Key Fields:** BENE_SEX_IDENT_CD (Gender), BENE_BIRTH_DT (Date of Birth), and SP_STATE_CODE (Location).
- **Relevance:** It also includes chronic condition indicators (e.g., Alzheimer's, Diabetes), which are critical predictors of patient risk and potential resource usage.

3.2 Inpatient Claims Data

This transactional dataset details specific hospital admissions and billing events.

- **Volume:** 66,773 rows and 81 columns.
- **Key Fields:** CLM_ID (Unique Claim Identifier), CLM_PMT_AMT (The target variable: Medicare Payment Amount), CLM_DRG_CD (DRG Code), and admission/discharge dates (CLM_FROM_DT, CLM_THRU_DT).
- **Relevance:** This file serves as the core training data for our regression model.

3.3 MS-DRG Reference Table

This metadata table provides clinical context for the billing codes found in the claims data.

- **Volume:** 747 rows and 9 columns.
- **Key Fields:** MS-DRG (The code matching the claims table), MS-DRG Title (Description of the condition), and Weights (The relative resource intensity of the condition).
- **Relevance:** DRG weights are the single most important predictor of payment, as Medicare reimburses based on the complexity of the case.

4. Methodology: Spark Implementation

Per the course requirements, all data wrangling, exploration, and machine learning were performed using **Apache Spark**. We utilized the PySpark API to handle the volume of data efficiently. The methodology followed a standard data science pipeline:

4.1 Data Ingestion and Preprocessing

The raw CSV files were loaded into Spark DataFrames. Given the distinct sources, our first step was to perform relational joins to create a unified dataset. We joined the Inpatient Claims data with the Beneficiary Summary using the DESYNPUF_ID key and subsequently joined the MS-DRG Reference table using the CLM_DRG_CD key.

Data cleaning steps included:

- **Casting Types:** Converting date strings (e.g., 20080101) into actual DateType objects to enable calculation.
- **Handling Nulls:** We assessed the dataset for missing values in critical columns like CLM_PMT_AMT and filtered out incomplete records to ensure model integrity.

4.2 Feature Engineering

To improve the predictive power of our model, we engineered several features:

- **Length of Stay (LOS):** We calculated the duration of hospitalization by subtracting CLM_FROM_DT from CLM_THRU_DT..
- **Categorical Encoding:** We utilized Spark MLlib's StringIndexer and OneHotEncoder to transform categorical variables such as BENE_SEX_IDENT_CD and SP_STATE_CODE into numerical vectors suitable for regression analysis.

- **Feature Vector Assembly:** All predictor variables (DRG Weight, LOS, Age, Gender, Chronic Conditions) were assembled into a single feature vector using the VectorAssembler.

4.3 Machine Learning Pipeline

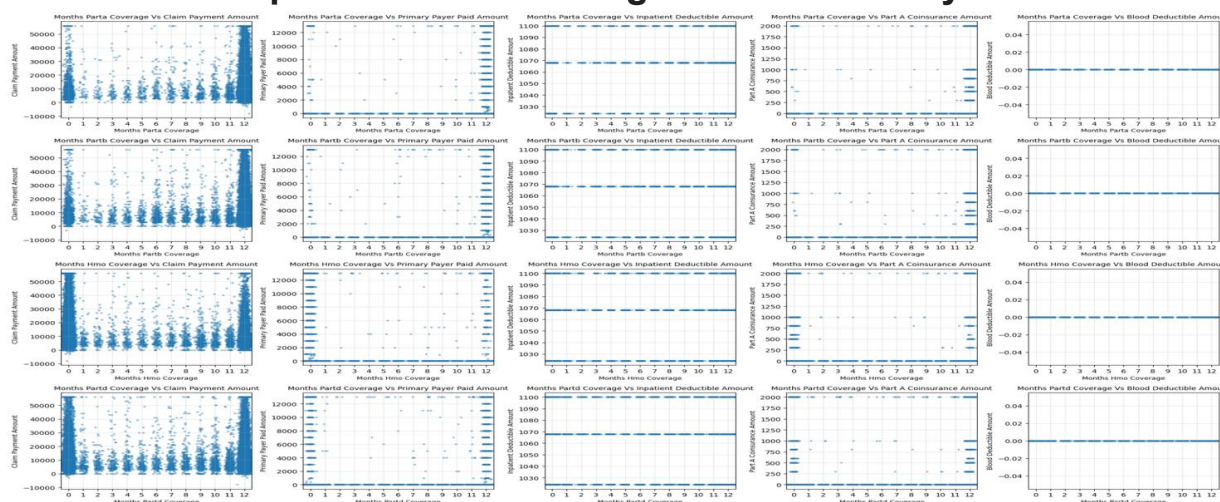
We selected a **Linear Regression** model for its interpretability and suitability for predicting continuous financial variables. We also experimented with random forest regressors to capture non-linear relationships.

- **Splitting:** The data was split into training (80%) and testing (20%) sets.
- **Training:** The model was trained to minimize the error between the predicted payment and the actual CLM_PMT_AMT.
- **Evaluation:** We evaluated the model using Root Mean Squared Error (RMSE) and R-Squared (R^2) metrics to assess accuracy.

5. Data Exploration and Insights

Before modeling, we conducted a preliminary data exploration to understand the underlying distributions and relationships in the data. This phase yielded three critical insights that shaped our modeling approach.

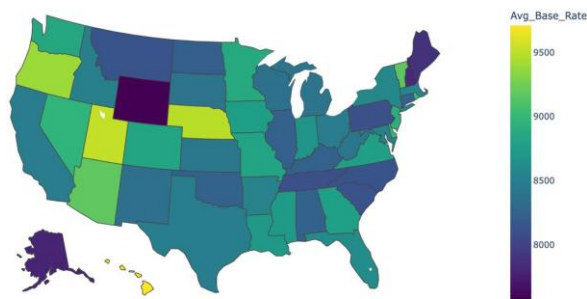
5.1 Relationship Between DRG Weight and Actual Payment



We visualized the relationship between the DRG Relative Weight and the Actual Medicare Payment. As hypothesized, the scatter plot confirmed a strong positive correlation: as the DRG weight (complexity) increases, the payment amount increases. This validates the use of DRG weight as a primary feature. However, we observed significant variance for lower-weight DRGs, where patients with similar complexity received widely different payments. This "vertical spread" in the data suggests that other factors-such as geographic adjustments or hospital-specific contracts-play a role, necessitating the inclusion of state and provider data in our model.

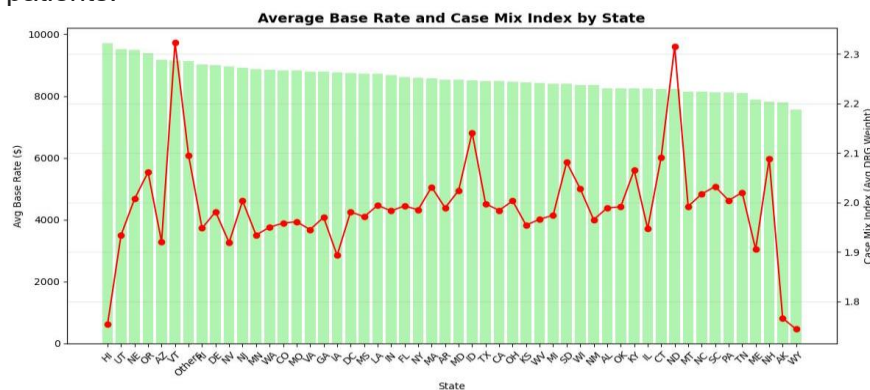
5.2 How Geography Influences Medicare Payments

Average Hospital Base Rate by State



Analysis of the Average Hospital Base Rate by State reveals distinct geographic disparities. Some states like Hawaii, Nebraska, and Utah have higher hospital payment rates compared to other states. This is often because Medicare provides additional funding to hospitals in locations where it is expensive to operate or difficult to secure staff, such as isolated or high-cost-of-living areas.

However, geography alone does not tell the whole story. To investigate further, we overlaid the Case Mix Index (CMI) to determine if high payments were strictly a result of treating sicker patients.

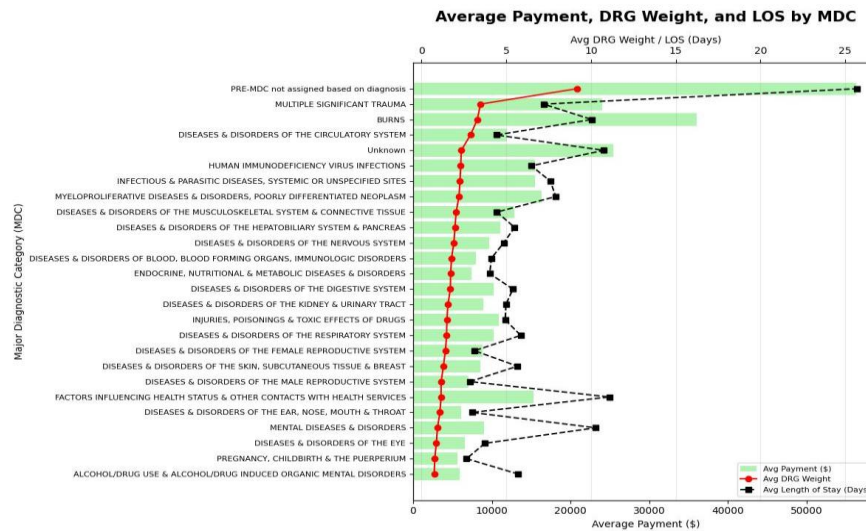


When we add the Case Mix Index (CMI)—which measures how severe or complex the patients' conditions are, we notice important nuances:

- High payments don't always mean hospitals treat more complex patients.
- For example, Vermont has very complex patients (high CMI) but only average payment rates.
- Conversely, Hawaii receives high payments even though its patient complexity is relatively low.

This confirms that hospital payments are not based solely on patient severity but are heavily influenced by location factors (cost of living, isolation, and wage indexes).

5.3 Analysis by Major Diagnostic Category (MDC)

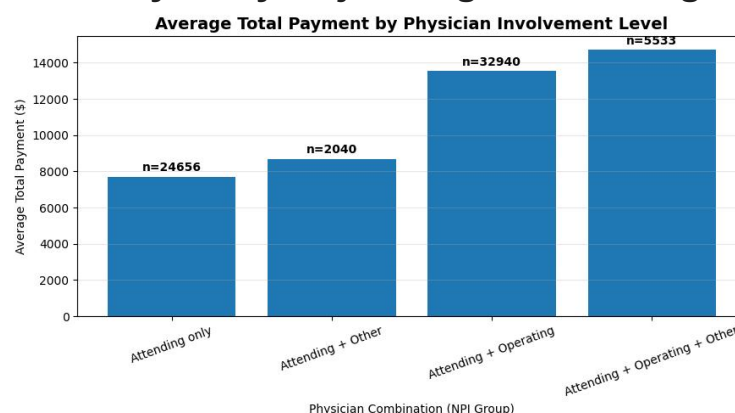


We aggregated the data to identify trends across Major Diagnostic Categories. Categories such as "Heart Failure," "Septicemia," and "Respiratory Infections" appeared frequently. Identifying these high-volume DRGs is crucial because even small systematic underpayments here can accumulate into millions in lost revenue.

Comparing Average Payment, DRG Weight (complexity), and Length of Stay (LOS), we found:

- **Higher complexity = higher payment:** Trauma, Burns, and major circulatory cases show high DRG weights and high payments due to intensive, resource-heavy care.
- **High payments without long stays:** Some categories (e.g., Burns, Circulatory Disorders) have high costs driven by specialized procedures rather than long hospitalization.
- **Long stays, low complexity:** Mental Health and Reproductive System cases stay longer but receive lower payments, reflecting lower-cost, non-surgical care.
- **Operational insight:** Categories like Infectious Disease show moderate complexity but high LOS, indicating opportunities to improve efficiency and reduce stay duration.

5.3 Analysis by Major Diagnostic Category (MDC)



The visualization confirms that claim costs generally rise with the **number of physicians involved**. Claims handled by only an attending physician show the lowest average payment, while those involving both attending and operating physicians show a significant increase.

The **highest costs are observed when all three NPIs** (attending, operating, and other) are present, suggesting that multi-physician involvement is a strong indicator of clinical complexity and higher resource utilization.

6. Results and Variance Analysis

The culmination of our project was the application of the trained Spark model to the test dataset to generate actionable business intelligence.

6.1 Variance Detection Logic

We applied the prediction model to the test set, generating a predicted_payment for every claim. We then calculated the variance percentage using the following formula:

$$\text{Variance}\% = \frac{\text{Actual} - \text{Predicted}}{\text{Predicted}} \times 100$$

Using a threshold of -10%, we created a binary flag. If Variance % < -10, the flag was set to 1 (Underpaid); otherwise, it was set to 0.

6.2 Interactive Dashboard Results

The output of our analyzer is structured as an interactive table for stakeholders.

- **Case Study 1 (Underpayment):** For Claim CLM123 at Crouse Hospital (DRG 290), our model predicted a payment of **\$12,500** based on the patient's complexity and stay duration. The actual payment received was **\$9,000**. The resulting variance was **-28%**. This claim was successfully flagged as "Underpaid" for immediate review.
- **Case Study 2 (Normal Payment):** For Claim CLM456 at Chastain Park Hospital (DRG 105), the predicted amount was **\$14,200** and the actual was **\$15,000**. With a variance of **+5.6%**, this claim was marked "Ok".

This result demonstrates the model's ability to filter noise and highlight only those claims that require human intervention, vastly increasing the efficiency of the revenue cycle team.

7. Problems Encountered

While the project was successful, we encountered specific challenges related to the nature of the data and the modeling complexity.

- **Synthetic Data Limitations:** Because we used DE-SynPUF data, some relationships were "noisier" than they would be in real administrative claims. The synthetic nature meant that certain columns occasionally lacked the tight logical consistency found in real-world billing,

which introduced error variance into our model.

- **Non-Linearity of Long Stays:** As identified in our data exploration, the relationship between payment and Length of Stay breaks down for long-term hospitalizations (over 25 days). Our primary Linear Regression model struggled to capture this "capping" effect, leading to higher prediction errors for outlier patients.
- **Data Size:** While Spark handled the data volume well, visualizing the results required aggregating or sampling the data, as plotting 66,000+ points directly in standard Python libraries (like Matplotlib) can be computationally expensive.

8. Conclusion

The **Claim Payment Variance Analyzer** successfully demonstrated the power of Big Data Analytics in solving a critical healthcare finance problem. By applying **Apache Spark** to process and model Medicare DE-SynPUF data, we achieved our goal of quantifying payment variance and detecting potential underpayments.

We moved beyond simple data storage to actionable inference, identifying that DRG weight and Length of Stay are the primary drivers of cost, while also revealing the financial risks associated with high-volume DRGs. The operational output of this project—a flagged list of underpaid claims—provides a clear, value-add solution for hospital administrators. This project highlights how predictive modeling can transform Revenue Cycle Management from a reactive, manual process into a proactive, data-driven strategy.

9. Works Cited

- Centers for Medicare & Medicaid Services. "Medicare Claims Synthetic Public Use Files (SynPUFs)." *CMS.gov*.
- Medical Group Management Association. "Stat: Insurers Underpay Providers by 7% to 11%." *MGMA*, 2022.
- American Hospital Association. "Fact Sheet: Medicare & Medicaid Underpayment." *AHA*, 2022.